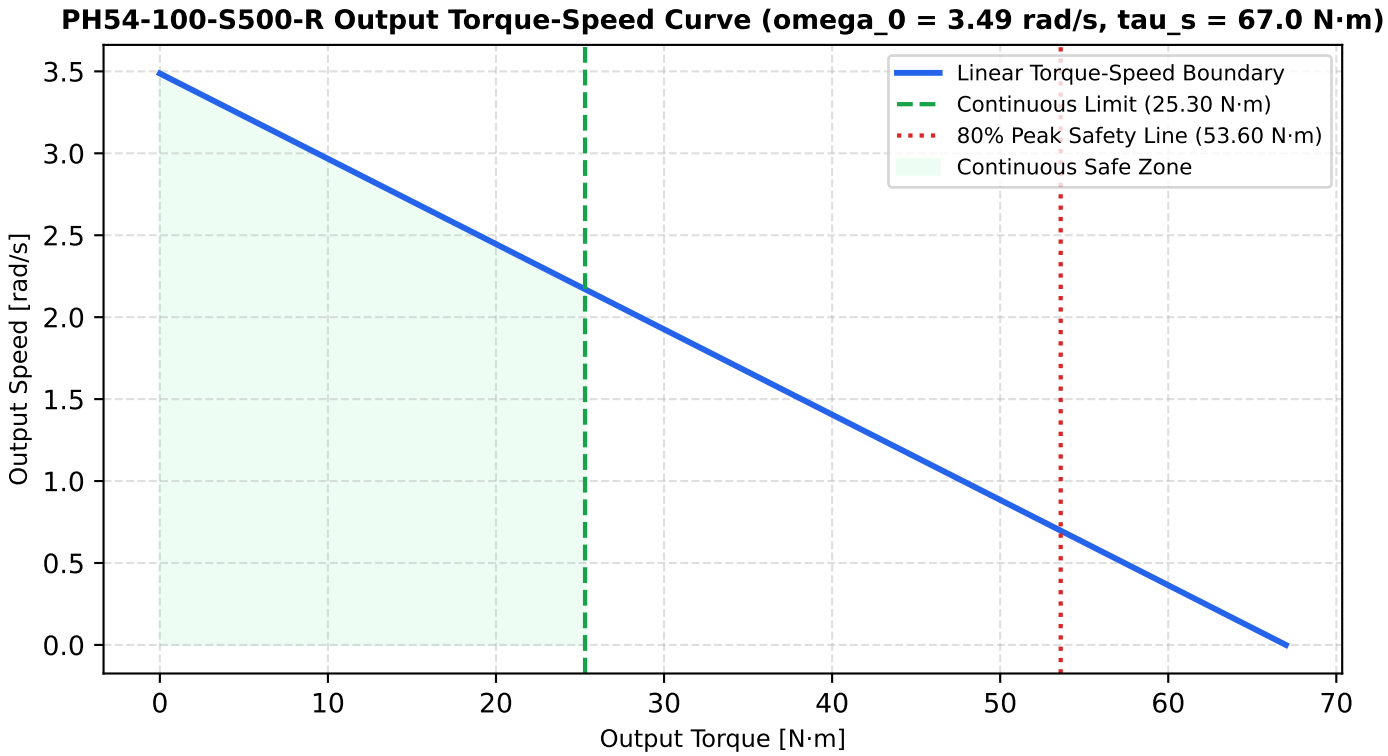
Part Number: PH54-100-S500-R

1. Electrical & Mechanical Specifications (Standard Reference)

Parameter	Symbol	Specification Value	Test Condition / Notes
Part Number	ID	PH54-100-S500-R	DYNAMIXEL-P
Manufacturer	Vendor	ROBOTIS	BLDC Servo
Rated Input Voltage	V_rated	24.0 V	Nominal DC Bus
Stall / Peak Torque	tau_stall	67.0 N.m peak/stall	At 24.0 V
Continuous Rated Torque	tau_cont	25.3 N.m rated continuous	25 degC ambient, continuous duty
No-Load Speed	omega_0	33.3 rpm (at 24.0V)	Unloaded output shaft
Actuator Mass (Weight)	m_act	740 g	Includes housing & internal gearhead
Internal Reduction	i_int	502.0 : 1 Cycloidal	Integrated gear stage
Envelope Dimensions	W x H x D	54.0 x 108.0 x 54.0 mm	Outer housing envelope



PH54-100-S500-R — Mechanical Drawing & Operating Guidelines

2. Mechanical Mounting & Frame Integration
   - Housing Envelope: 54.0 mm x 108.0 mm x 54.0 mm
   - Total Verified Mass: 740 g (0.7400 kg)
   - Mounting Pattern: M2.5 / M3 ISO radial & horn flange threads

3. Thermal & Duty-Cycle Notes
   - Continuous operation must remain at or below the continuous rated torque.
   - If continuous torque is not explicitly tabulated on Page 1, apply a 0.33x stall torque
   - Peak torque operation (>80% of available speed-dependent torque) risks step loss or the

4. Citation Verification Hash
   - Part ID: PH54-100-S500-R | Source Document: ph54-100-s500-r.pdf | Spec Table: Page 1